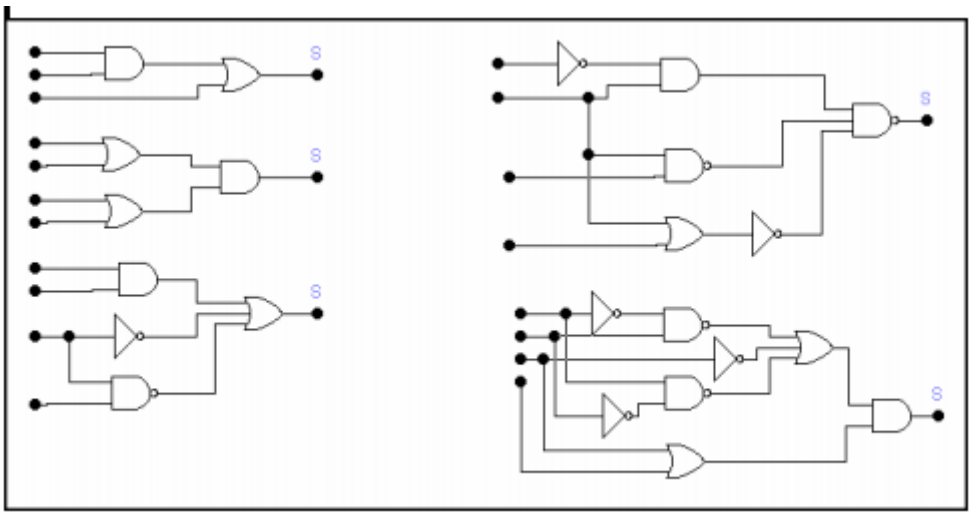
Exercícios

1. Escrever as expressões lógicas dos circuitos apresentados abaixo:



a) A*B + C = S
b) (A+B) * (A+C) = S
c) ((A*B) + NOT C) + NOT (C * D)
d) NOT((NOT A*B) * (NOT (B*C) * NOT (B+D))) = S
e) (NOT (NOT A*B) + NOT C + NOT (A* NOT B)) * (C+D) = S

2. Utilizando o logism desenhe os circuitos resultantes das expressões lógicas apresentadas:

a) $S = (A+B).C.(B+D)$

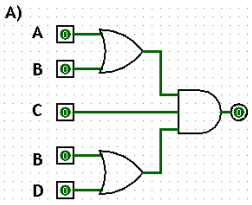
b) $S = A.B.C + (A+B).C$

c) $S = (\overline{\overline{A.B} + \overline{C.D}})$

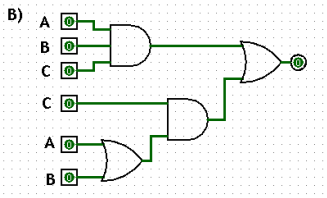
d) $S = [(\overline{\overline{A} + \overline{B}}) + (\overline{\overline{C}.D})].\overline{D}$

e) $S = [(\overline{\overline{A.B}}) + (\overline{C.\overline{D}})].E + [(A.\overline{D}.\overline{E}) + (C.D.E)].\overline{A}$

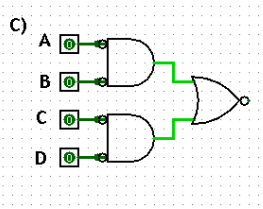
a)



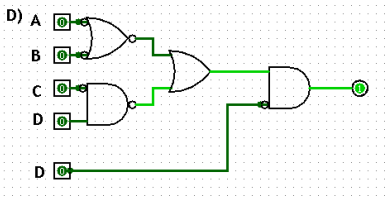
b)



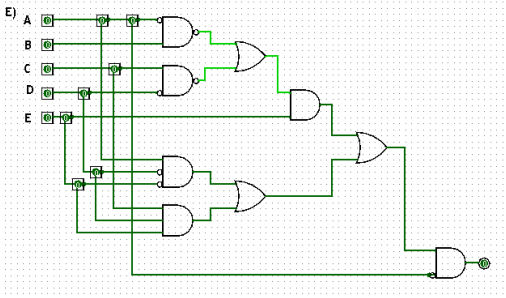
c)



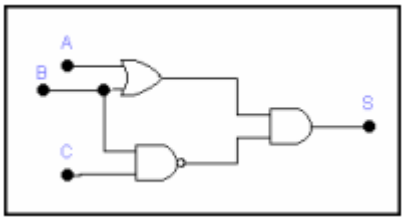
d)



e)



3. Obtenha a tabela verdade e a expressão lógica através do circuito



RESPOSTA:

(A+B) * NOT (B*C) = S

TABELA VERDADE:

A	B	C	NAND	S
0	0	0	1	0
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0